Influence of Popliteal Vein Thrombosis on Subsequent Ambulatory Venous Function Measured by Near-infrared Spectroscopy

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BACKGROUND: To determine the influence of the site affected by thrombi on the subsequent venous physiology, we examined patients with postthrombotic syndrome (PTS) with respect to ambulatory venous function using near-infrared spectroscopy (NIRS).

METHODS: Fifty-one limbs of 45 patients, for whom more than 1 year had passed since an acute episode of deep vein thrombosis, were studied. Seventeen limbs were asymptomatic, 27 had mild symptoms (edema only), and 7 showed severe symptoms (skin changes). The mean duration of PTS was 8.2 years. All of the patients underwent a treadmill walking test with simultaneous NIRS. Deoxygenated hemoglobin was continuously measured during exercise. The ambulatory venous retention index (AVRI) obtained from the serial deoxygenated hemoglobin changes was calculated in each patient. The location of thrombi at the onset of deep vein thrombosis was identified by venography.

RESULTS: The calculated AVRI was apparently related to the clinical symptoms of PTS. The limbs initially involved with popliteal vein thrombosis showed significantly higher AVRI values than those without popliteal vein thrombosis.

CONCLUSIONS: The clinical severity of PTS is correlated well with the degree of venous retention during exercise. Initial involvement of the popliteal vein is an important factor determining subsequent venous hemodynamics in patients with PTS. *Am J Surg.* 1999;177:111–116. © 1999 by Excerpta Medica, Inc.

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The majority of patients who have an episode of deep vein thrombosis (DVT) are at risk of developing the postthrombotic syndrome (PTS), which may lead to considerable long-term morbidity. The pathophysiologic mechanisms of PTS consist of persistent outflow obstruction due to residual organized thrombus and venous reflux caused by valvular destruction, with one or the other being physiologically dominant. Both of these conditions produce venous retention and result in the clinical manifestations of PTS, such as ulceration, chronic swelling, and skin changes.

Duplex scanning can evaluate the degree of reflux or obstruction in identified individual veins, and has revealed the rate of spontaneous lysis of thrombi[1] and the incidence of venous reflux after DVT.[2,3] It has also been well documented that there is a relationship between the development of PTS and the anatomic and functional changes in the venous system, using duplex scanning.[4–6] However, it has not been clearly defined whether the site of the affected vein influences subsequent ambulatory venous function in PTS, mainly due to lack of appropriate methods. Near-infrared spectroscopy (NIRS) is a new modality that can be used to evaluate venous retention during walking noninvasively.[7] NIRS is able to assess ambulatory venous function in the whole leg, in spite of whether reflux or obstruction is pathophysiologically dominant. The aim of this study was to correlate the degree of venous retention during walking with the sequelae of clinical symptoms, and to determine the influence of the site initially affected by thrombi on the subsequent degree of ambulatory venous dysfunction in patients with PTS, by using NIRS.

MATERIALS AND METHODS

From January 1995 to March 1997, 51 limbs of 45 consecutive outpatients who were referred to our vascular laboratory with the suspected diagnosis of PTS were included in our study. For the purpose of this study, we excluded patients if one or more of the following conditions were present: (1) patients in whom the diagnosis of DVT was not verified by venography at the onset; (2) patients for whom less than 1 year had passed since an acute episode of DVT; (3) patients with a history of symptomatic recurrent DVT in the affected limb; (4) preexisting symptoms caused by chronic venous insufficiency before the onset of DVT; and (5) physical disability making walking impossible.

The patients consisted of 23 men and 22 women with a mean age of 54.6 years (range 17 to 81). The mean duration of PTS was 8.2 years (range 1 to 33).

In all patients, clinical severity was assessed by the same

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doctor at the time of their visit. The clinical examination focused especially on the signs and symptoms of postthrombotic sequelae such as swelling, pigmentation, and ulceration. The limbs were classified according to the clinical severity of their condition following the guidelines in *Reporting Standards in Venous Disease: An Update* proposed by an International Consensus Committee associated with the American Venous Forum in 1995:[8] class 0, no visible or palpable signs of venous disease; class 1, telangiectases, reticular veins, or malleolar flare; class 2, varicose veins; class 3, edema without skin changes; class 4, skin changes such as pigmentation or lipodermatosclerosis; class 5, skin changes with healed ulceration; and class 6, skin changes with active ulceration. For practical purposes, the patients were divided into three groups: asymptomatic group consisting of patients with clinical severity class 0; mild group consisting of patients with class 1 to 3; and severe group consisting of patients with class 4 to 6. This major division is based on the presence or absence of symptoms or skin changes. In all of the patients, clinical examination provided the following distribution: asymptomatic group, 17 limbs; mild group (edema only), 27 limbs; severe group (skin changes), 7 limbs. No patients had ischemic symptoms associated with arterial disease, and all had either palpable pedal pulses or an ankle brachial pressure index greater than or equal to 1.0.

The location and extent of thrombi at the onset of DVT were identified by venography. For practical purposes, the veins of the lower limbs were divided into four segments: calf veins, below the tibio-peroneal junction; popliteal vein; femoral veins, including superficial and deep femoral veins; and iliac veins, including common femoral and iliac veins. The presence or absence of a lesion in each segment was examined in all of the patients.

To assess the ambulatory venous function, all patients underwent a treadmill walking test with simultaneous NIRS. The details of this technique have been described previously.[7] Briefly, the optrodes of a near-infrared spectrometer (Shimadzu Co., Non Invasive Oxygenation Monitor OM-100A, Japan) were positioned on the posterior aspect of the calf. The light guide and the near-infrared detector were applied to the skin 4 cm apart longitudinally with a bandage. Then, patients performed a treadmill walking test, on a gradient of 12% at a speed of 2.4 km/h. Exercise time was 5 minutes. Changes in deoxygenated hemoglobin (DeoHb) were continuously measured by NIRS during exercise. The ambulatory venous retention index (AVRI) obtained from the serial DeoHb changes was calculated in each patient as a parameter of venous function during walking **(Figure 1)**.

Compression stockings were adapted by experienced doctors and prescribed for each patient. The patients were instructed to wear the elastic stockings during daily activities and informed about the importance of compression therapy. The compliance with compression therapy was assessed at the time of their visit. Compliance was defined as regular when the patients claimed that they wore the elastic stockings on the average of at least 5 days per week. It was considered sporadic when the stockings were applied for less than 5 days.

The statistical methods used in this paper included calculation of mean ± standard deviation. Differences between means of NIRS data were tested by analysis of variance (ANOVA) with post hoc testing by Scheffe's F test because the number of available data in each group was different. The Mann-Whitney *U* test or chi-square test was applied when appropriate. All measurements were made after obtaining informed consent from the patients.

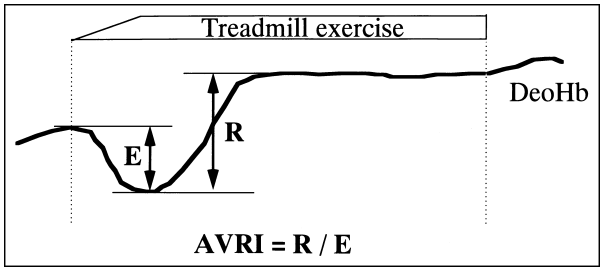


Figure 1. Definition of ambulatory venous retention index (AVRI). AVRI = rise from the bottom to the new plateau level (R)/fall from the baseline to the bottom (E). The less the expulsion during exercise (E), the more venous reflux or retention (R), and therefore, the greater AVRI, which indicates a higher severity of ambulatory venous insufficiency.

RESULTS

Acute DVT affected the right leg in 13 patients, the left leg in 26 patients, and was bilateral in 6 patients. Features of PTS were present in 34 (67%) of the 51 affected limbs, and 17 (33%) limbs were asymptomatic. In the severe group (class 4 to 6), overt ulceration was found in only 2 limbs during the follow-up period.

The location and extent of initial thrombosis are listed in **Table I**. Sixteen limbs (31%) had involvement of a single venous segment and 35 limbs (69%) had thrombosis at two or more segments. After multisegment venous thrombosis, 27 out of 35 limbs (77%) had some signs or symptoms of PTS, whereas postthrombotic changes were found in 44% (7 of 16) of the limbs with single involvement. There was a significant difference in the occurrence of subsequent symptoms according to the initial extent of thrombi ($P < 0.03$, Fisher's exact test). Especially, as for limbs with isolated calf vein thrombosis, 7 of 11 limbs (64%) were asymptomatic and no trophic skin changes were found during the follow-up period. With regard to the location of DVT, the incidence of subsequent clinical symptoms was significantly higher in limbs with thrombosis involving the popliteal vein, as compared with limbs in which the popliteal vein was not involved (21 of 25 versus 13 of 26; $P < 0.02$, Fisher's exact test). Furthermore, 6 of 7 patients in the severe group had popliteal involvement. On the other hand, there were no significant differences in the other segments (calf, femoral and iliac).

The results of the calculated value of AVRI obtained from NIRS measurements were stratified according to the clinical severity of venous disease and are shown in **Table II**. AVRI studies revealed significant differences between the three groups, analyzed by ANOVA ($P < 0.0001$). Between two groups, the mean AVRI of the asymptomatic group was significantly lower than that of the mild group (0.98 ± 0.36 and 1.94 ± 0.93, respectively; $P = 0.003$). Moreover, there was a significant difference between the

TABLE I
Localization of Initial Thrombosis Identified by Venography

Site of Thrombi	Number of Limbs
Single segment	
Iliac	1
Femoral	3
Popliteal	1
Calf	11
Multisegment	
Iliac + femoral	6
Iliac + calf	2
Femoral + popliteal	1
Femoral + calf	2
Iliac + femoral + popliteal	4
Iliac + femoral + calf	1
Iliac + popliteal + calf	1
Femoral + popliteal + calf	14
All segments affected	4

TABLE II
Ambulatory Venous Retention Index versus Clinical Severity

Clinical Severity	Number of Limbs	AVRI (Mean ± SD)
Asymptomatic	17	0.98 ± 0.36*
Mild	27	1.94 ± 0.93†
Severe	7	3.10 ± 1.28

AVRI = ambulatory venous retention index.
ANOVA; P *<0.0001.*
** Asymptomatic versus mild,* P *= 0.003.*
† Mild versus severe, P *= 0.008.*

mild group and the severe group (1.94 ± 0.93 and 3.10 ± 1.28, respectively; $P < 0.01$).

When AVRI was compared according to the initial extent of thrombi, limbs with single involvement showed significantly lower AVRI values than those with multisegment involvement (1.21 ± 0.51 and 2.04 ± 1.17, respectively; $P = 0.005$, Mann-Whitney *U* test). Regarding the relationship between the location of initial thrombi and the subsequent ambulatory venous function, the presence or absence of involvement in each venous segment and corresponding AVRI values are depicted in **Figure 2**. Among all segments with and without involvement, the mean AVRI value was highest in the limbs with popliteal vein involvement. When the mean AVRI values were compared between the limbs with and without initial thrombi in each segment, the limbs with initial involvement showed significantly higher AVRI values than those without involvement in the popliteal and femoral segments ($P < 0.0001$ and $P < 0.02$, respectively, Mann-Whitney *U* test). On the other hand, the mean AVRI values were not substantially different between limbs with and without involvement of the calf or iliac segment. Moreover, among the limbs that had femoral vein thrombi at the onset of DVT, AVRI data were analyzed in detail in order to examine the influence of concomitant popliteal vein thrombi. There were 35 limbs with femoral vein involvement; 23 had coexisting popliteal vein involvement and 12 did not. The mean AVRI value of the limbs with popliteal vein involvement was significantly higher than that of the limbs without popliteal vein involvement (2.31 ± 1.11 and 1.27 ± 0.39, respectively; $P = 0.0005$, Mann-Whitney *U* test; **Figure 3**). Among the limbs with popliteal vein involvement, statistical analysis of the influence of concomitant femoral vein involvement could not be performed, because almost all (23 of 25) limbs had simultaneous femoral vein thrombosis.

As for compression therapy, compliance was regarded as regular in 29 (57%) limbs, and 22 (43%) limbs had sporadic compliance or no compression. The patients who wore elastic stockings regularly demonstrated significantly lower AVRI values than those who used stockings only occasionally (1.41 ± 0.65 and 2.27 ± 1.33, respectively; $P = 0.005$, Mann-Whitney *U* test).

COMMENTS

The long-term clinical outcome of patients after an episode of DVT is not identified with certainty. Early studies[9,10] have reported that more than 80% of patients with a clinical diagnosis of DVT had stasis ulceration during a 15-year follow-up period. In recent years, advances in the treatment of this disease such as anticoagulation and compression therapy appear to have reduced the prevalence of severe postthrombotic complications. Nevertheless, the problem of PTS has not eliminated and a substantial number of patients continue to suffer from less severe forms of this disorder.

Despite the considerable prevalence of PTS in clinical practice, objective assessment of the severity has been difficult. Recently, it has been recognized that duplex scanning provides useful information about the pathophysiology of PTS. It can evaluate the degree of venous reflux as well as residual obstruction in identified individual veins, and the true incidence, natural history, and long-term consequence of DVT have been ascertained by the application of this method. Although duplex scanning is highly sensitive in detecting anatomic and functional abnormalities in the specific veins examined, evaluation of the overall degree of venous insufficiency is not possible. Additionally, duplex imaging is difficult to obtain under the condition of exercise, such as walking, which can offer more physiologic information. In contrast, NIRS is capable of noninvasive and continuous monitoring of tissue oxygenation and the changes in venous blood volume even during exercise. NIRS has the great advantage of being able to be used in dynamic conditions regardless of the patient's movement, and to provide valuable information on ambulatory venous function. In this regard, NIRS is thought to be an essentially different modality from duplex scanning.

Plethysmography is another frequently used technique to investigate venous function in patients with PTS. However, assessment of the physiological effect of residual obstruction in the chronic phase is still difficult with this method. Several authors have reported that maximal venous outflow was correlated poorly with the presence of postthrombotic changes, due to collateral formation or recanalization.[11,12] Air-plethysmography (APG) is considered to be the most established and reliable modality that provides several useful parameters concerning venous hemodynamics.[13] However, the venous filling index obtained from APG tends to underestimate the degree of venous reflux in patients with proximal venous obstruction.[14] It

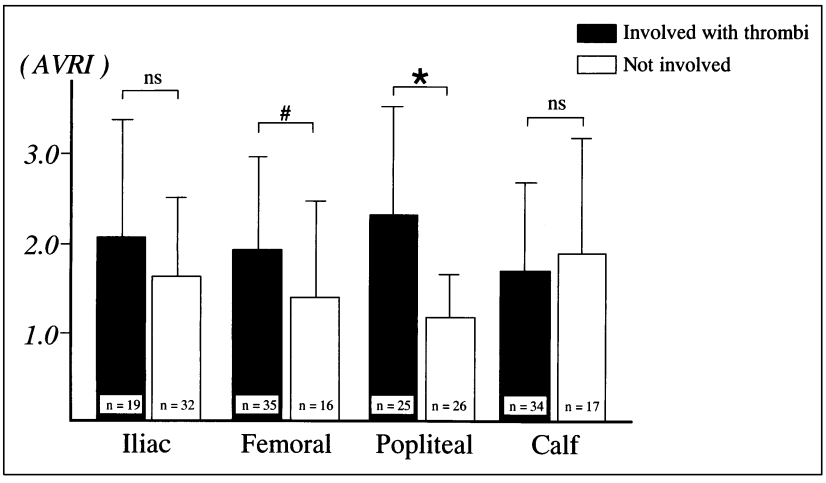


Figure 2. Mean ambulatory venous retention index values for all segments, grouped according to the presence of initial thrombi. Error bars show standard deviation. NS = no statistical significance. *P <0.0001, Mann-Whitney U test. #P <0.02, Mann-Whitney U test.

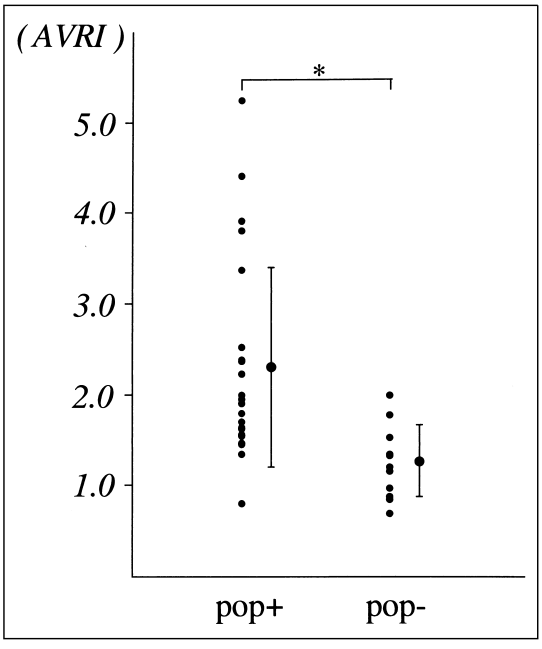


Figure 3. Ambulatory venous retention index values in limbs with femoral vein involvement, in those with and without concomitant popliteal vein involvement (pop+ = popliteal vein was initially involved; pop− = popliteal vein was not involved). Data from individual limbs, with mean and standard deviation, are shown. *P = 0.0005, Mann-Whitney U test.

has also been reported that there is a poor correlation between the residual volume fraction and the ambulatory venous pressure when residual obstruction is present.[15] The lack of an appropriate method to evaluate venous function under the simultaneous existence of both reflux and obstruction remains an important issue.

In consideration of these facts, we hypothesized that treadmill walking exercise could provoke venous retention in the postthrombotic state, whether the pathophysiology was obstruction or reflux or both. Therefore, we applied NIRS to patients with PTS and evaluated ambulatory venous retention by measuring the changes in DeoHb during exercise. NIRS in combination with treadmill test can assess venous hemodynamics during walking, even under the condition in which residual obstruction and venous reflux coexist, and will shed light on ambulatory venous physiology in patients with PTS, which was previously poorly defined.

In the current study, we observed a relatively low incidence of severe PTS after acute DVT. Only 2 (4%) out of 51 limbs in our series developed a venous ulcer, and class 4 skin changes were present in 5 limbs (10%) at a mean follow-up of 8 years. These data are in concordance with other recent reports regarding the prevalence of postthrombotic symptoms.[4,16,17] Our data also demonstrated a clear correlation between the incidence of subsequent symptoms and the site or extent of initial thrombi. PTS occurred more frequently in limbs with multisegment involvement than in those with single involvement. Moreover, popliteal vein involvement was found to be a factor of paramount importance in determining which patients with DVT will develop PTS signs later. These findings in this study confirmed several earlier observations.[18,19]

Of greatest interest in the present study is the ambulatory venous dysfunction in postthrombotic patients. We used AVRI obtained from NIRS measurements as a parameter for evaluating the venous hemodynamics of PTS. The usefulness of this parameter in the assessment of venous insufficiency has been verified previously.[7] AVRI showed trends indicating the progressive worsening of venous hemodynamics as the symptoms progressed from asymptomatic to mild PTS and from mild to severe PTS. There have been many studies[11,18] focusing on the correlation worsening of the clinical severity of PTS with hemodynamic venous deterioration in terms of plethysmographic examination, but few studies could discriminate a hemodynamic difference between mild and severe PTS. In this regard, our results indicated that NIRS measurements were more reliable than other modalities in distinguishing the clinical severity of PTS. A strong correlation between AVRI and the severity was evidenced, which can be considered a valuable parameter in the assessment of PTS. However, the

number of class 4 to 6 patients was so small that further accumulation of data will be necessary.

AVRI studies have revealed a difference in the ambulatory venous function according to the extent of initial thrombi; limbs with multisegment involvement had significantly higher AVRI values than those with single involvement. It is conceivable that a greater extent of thrombi at the onset of DVT results in more marked deterioration of consequent venous physiology. However, the hemodynamic difference between limbs with single and multisegment involvement observed in our patients can arise from the disparity in the distribution of popliteal vein involvement. In fact, popliteal involvement was more prevalent in the multisegment group than in the single group (24 of 35 versus 1 of 16; $P < 0.01$, Fisher's exact test). Furthermore, excluding the patients with popliteal involvement, there was no significant difference between these two groups (1.22 ± 0.44 and 1.23 ± 0.52, respectively; $P = 0.94$, Mann-Whitney U test). These results suggest that the site of initial thrombi rather than the extent may play a more important role in determining subsequent venous hemodynamics.

In this study, we intended to elucidate the correlation between the site initially affected by thrombi and the subsequent degree of impairment of venous function. As a result, we found that popliteal and femoral vein involvement led to significant abnormal venous retention during exercise. Furthermore, analyzing the patients with femoral vein involvement, concomitant popliteal vein involvement was closely associated with worsening of ambulatory venous function. These findings of our study clearly demonstrate that initial popliteal vein involvement is the most important factor causing later deterioration of venous hemodynamics in patients with PTS. These results are in accord with previous reports,[20,21] which have emphasized the significance of the popliteal segment in the pathophysiology of PTS. It has been suggested that the popliteal vein valve acts as a gatekeeper for the calf muscle pump.[22,23] Although several authors stressed the significant correlation between the status of the popliteal vein and the subsequent development of clinical symptoms, only a few studies addressed the overall venous function in limbs with PTS. As a functional study, Shull et al[20] found that popliteal vein reflux was the most important factor determining the ambulatory venous pressure. However, their study only included patients with iliofemoral or femoral thrombosis, so the effect of initial popliteal vein involvement could not be defined. From this point of view, our study is thought to include novel findings, which demonstrate the relationship between the site initially affected by thrombosis and the subsequent degree of ambulatory venous dysfunction.

Additionally, there have also been reports[5,22,24] referring to the significant role of distal (popliteal/calf) deep veins to prevent postthrombotic sequelae. Nevertheless, according to our observation, consequent ambulatory venous function assessed by AVRI was not affected by the presence of initial calf vein thrombi. Moreover, limbs with calf vein thrombosis in our series showed a benign long-term clinical course, which was in agreement with other investigators.[17,25] This difference may arise from analyzing popliteal and calf veins together in other studies. Since they have not distinguished the popliteal from the calf vein segment, it is unclear which segment is the predominant factor in determining the development of PTS. Another explanation for this discrepancy might reside in the natural course of calf vein thrombi. Recently, it has been recognized by meticulous duplex examination that a smaller proportion of veins in the calf region developed venous reflux.[2,3] The inherent property of the calf veins, ie, the greatest concentration of valves and their small caliber, which causes thrombus fragmentation and plasminogen activation, may contribute to valve preservation in this segment.[3,26] On the other hand, once reflux in the calf vein occurs, the ankle level would be subject to a long uninterrupted column of blood, which produces higher hydrostatic pressure and may result in the development of PTS.[22,24] The abovementioned authors, who emphasized the importance of distal veins, have mainly investigated the correlation between the presence of venous reflux in calf veins at the chronic phase of DVT and subsequent symptoms. However, this seems to be another issue because the purpose of the current study was focused on how the initial thrombosis affected clinical sequelae and later venous hemodynamics. Calf deep vein thrombosis still remains controversial in terms of its clinical importance. At any rate, the data obtained from our study cast doubt on the important role attributed to calf vein thrombosis in the development of postthrombotic sequelae, but further investigation will be needed on this issue.

Compression therapy with elastic stockings presently represents the only proven effective treatment available for patients with PTS. Although our study was not designed to confirm the efficacy of compression therapy in a randomized fashion, the positive effect of this treatment is obvious, comparing the clinical outcome of the compliant and noncompliant patients in our series. From the hemodynamic aspect, patients with regular compression therapy showed significantly better function during walking than those whose compliance was classified as sporadic. In this study, we found that there were some patients who showed normal AVRI values, which indicated that they had normal ambulatory venous function. It may be reasonable for these particular patients not to wear elastic stockings, with their expense, discomfort, and potentially poor patient compliance. However, it is considered necessary to repeat venous vascular laboratory evaluation at scheduled intervals because delayed venous reflux or rethrombosis may occur.[26,27]

We excluded patients for whom less than 1 year had passed since the onset of DVT, because venous hemodynamics are thought to fluctuate during this period. Actually, rapid hemodynamic changes in the venous system due to collateral formation, recanalization, and valvular destruction occur during the early phase after DVT. Under such dynamic conditions, it is difficult to accurately assess the influence of initial thrombosis on consequent venous function. Concerning the duration of this dynamic state, Meissner et al[26] elucidated with the use of duplex scanning that the vast majority of veins affected by thrombi in each segment had completely recanalized within 1 year, with or without reflux. Therefore, we regarded the patients for whom 1 year had passed from the onset of DVT as having

relatively stable venous physiology, although there must be differences between individuals. We also excluded patients with recurrent DVT during the follow-up period because new thrombi have a possibility of altering the effect of initial thrombi upon venous hemodynamics.

We conclude that the clinical severity of PTS is correlated well with the degree of venous retention during walking exercise, and that initial involvement of the popliteal vein is the most important factor influencing subsequent venous hemodynamics in patients with PTS. We believe that NIRS will shed light on venous hemodynamics during exercise in chronic venous disease which was previously poorly understood, and ultimately lead to better patient care.

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